

ONESOL TECH HOTEL

AI Receptionist & Operations Dashboard

Project Summary — March 2026

SECTION 1 WHAT HAS BEEN BUILT

1. WhatsApp AI Receptionist (Bot)

- Live on WhatsApp via Green API
- Powered by OpenAI GPT-4o-mini
- Bot persona: Alex — Senior Front Desk Executive
- Responds to guests 24/7 in natural, human-like language

2. Intelligent Conversation Engine

- Full conversation memory — remembers room number, previous requests, and guest context throughout the chat
- Smart priority routing:
 - Greetings → menu on first contact, brief acknowledgment on return
 - Menu shortcuts (1–16) → instant replies
 - Service keywords → confirm + ask room number → log request
 - Free-form questions → GPT with full conversation history
- Context protection: if the bot asks a question, the next reply is never misread as a menu command
- First contact vs returning guest handled differently

3. Human-Like Response System

- Alex speaks like a real front desk executive — warm, calm, solution-focused
- Emotional intelligence layer:
 - Tired guest → brief, fast, no pleasantries
 - Upset guest → apologise first, then solve
 - Happy guest → match their warmth, light-hearted
 - Rude guest → calm, professional, never defensive
- Time-aware greetings (good morning / afternoon / evening / night)
- Natural confirmations: "Certainly! I'll have that with you shortly" instead of "Request received."
- Handles absurd requests (e.g. 1000 pillows) with warm humour

4. Service Request Management

- Detects service requests from natural language (pillows, food, taxi, AC repair, etc.)

- Automatically categorises into departments: Housekeeping, Room Service, Concierge, Maintenance
- Asks for room number before confirming, then logs the request to the dashboard
- Quantity limits enforced by room type:

Item	Standard	Deluxe	Executive Suite	Family Suite
Extra Pillows	2 max	2 max	2 per guest	4 max
Extra Towels	2 max	2 max	2 per guest	4 max
Toiletries	1 set	1 set	1 per guest	2 sets
Bedsheets	1 change	1 set	1 set	1 set

- Misuse prevention:
 - Executive Suite: bot always asks number of guests, calculates max = 2 × guests
 - Repeated quantity increases: bot holds firm politely after 2nd attempt beyond limit
 - Absurd quantities (50+): warm humour response + enforces room limit

5. Real-Time Operations Dashboard

- Runs at localhost:3000 alongside the bot — no separate server needed
- Live data via WebSocket (Socket.io) — updates the moment a WhatsApp message arrives
- 7 fully-built pages:

Page	What It Shows
Dashboard	6 live KPI cards, weekly chart, dept chart, status pie, response time trend
Conversations	Live chat list, full chat view, Take Over & Escalate controls
Requests	Full service request log with department and status tracking
Departments	Configurable load limits and auto-escalation rules per department
Analytics	AI accuracy, peak hours bar chart, escalation reasons, AI vs Human rate
Staff	Staff directory with online/offline status and performance stats
Settings	SLA threshold, VIP tagging, automation level, integration status

SECTION 2 CURRENT TECH STACK

Layer	Technology
Backend	Node.js, Express, Socket.io
AI Engine	OpenAI GPT-4o-mini
WhatsApp	Green API
Dashboard UI	React, Vite, Tailwind CSS, Recharts
Data Storage	In-memory (session-based, resets on restart)
Deployment	Local (localhost:3000)

SECTION 3 NEXT STEPS

Priority 1 — Core Stability

- Persistent database (MongoDB / PostgreSQL) — all data currently resets on server restart
- Guest name capture and storage across sessions

- Multi-language support (Urdu, Arabic, etc.)

Priority 2 — Operations

- Staff can reply to guests directly from the dashboard
- Real escalation workflow — assign requests to specific staff members
- Push notifications to staff phones when a new request arrives
- SLA timer — auto-flag requests not actioned within X minutes

Priority 3 — Guest Experience

- VIP guest tagging — different tone and limits for VIP rooms
- Proactive messaging — bot sends reminders (e.g. check-out time reminder at 10 AM)
- Order tracking — guest can ask "what's the status of my order?"
- Feedback collection at end of stay

Priority 4 — Business Intelligence

- Export reports (PDF / Excel) from the dashboard
- Historical analytics (week-over-week, month-over-month)
- Guest satisfaction scoring

Priority 5 — Scale

- Multi-property support (multiple hotels on one dashboard)
- Admin roles (Manager vs Staff vs Viewer access levels)
- API integration with hotel PMS (Property Management System)

SECTION 4 HOW TO RUN

Start the bot and dashboard with a single command:

```
node server.js
```

- WhatsApp webhook → <http://localhost:3000/webhook>
- Dashboard → <http://localhost:3000>
- API stats → <http://localhost:3000/api/stats>